



6388 - Diagnosing Residual Correlated Noise in Small Planet Transmission Observations for Maximum Feature Detection

Cycle: 3, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Johanna Teske (PI)	Carnegie Institution of Washington
Dr. Nicole L. Wallack (CoI) (CoPI)	Carnegie Institution of Washington
Dr. Luis Welbanks (CoI) (CoPI)	Arizona State University
Dr. Peter McGill (CoI)	Lawrence Livermore National Laboratory

OBSERVATIONS

ABSTRACT

Do 1-3 Rearth exoplanets have atmospheres, what are they composed of, how similar or different are they from Solar System planets? These questions are driving hundreds of hours of JWST observations, but so far the results have been inconclusive or pose challenges to interpretation. Examining these results suggests that there is likely correlated noise still lurking in the observations that may be limiting the precision and thus atmospheric detection significance in small planets, jeopardizing our ability to answer the questions above. Here we propose an archival program encompassing all Cycle 1 and 2 NIRSpec/G395H transmission observations of 1-3 Rearth planets to investigate the correlated noise properties in these data. We will uniformly reduce all of the observations and measure the correlated noise, apply two different correlated noise models (one parametric and one non-parametric), compare the performance of these models and pinpoint where they are making the most difference, and assess the physical origins of the correlated noise across detectors, visits, and our full sample of 28 small planets orbiting a variety of stellar types and magnitudes. From our lessons learned, we will provide the community with valuable recommendations for future design and analysis of NIRSpec/G395H observations. We will also provide updated spectra of the small planets that have reduced if not mitigated correlated noise, which can be used for higher-significance atmospheric feature detection.

OBSERVING DESCRIPTION